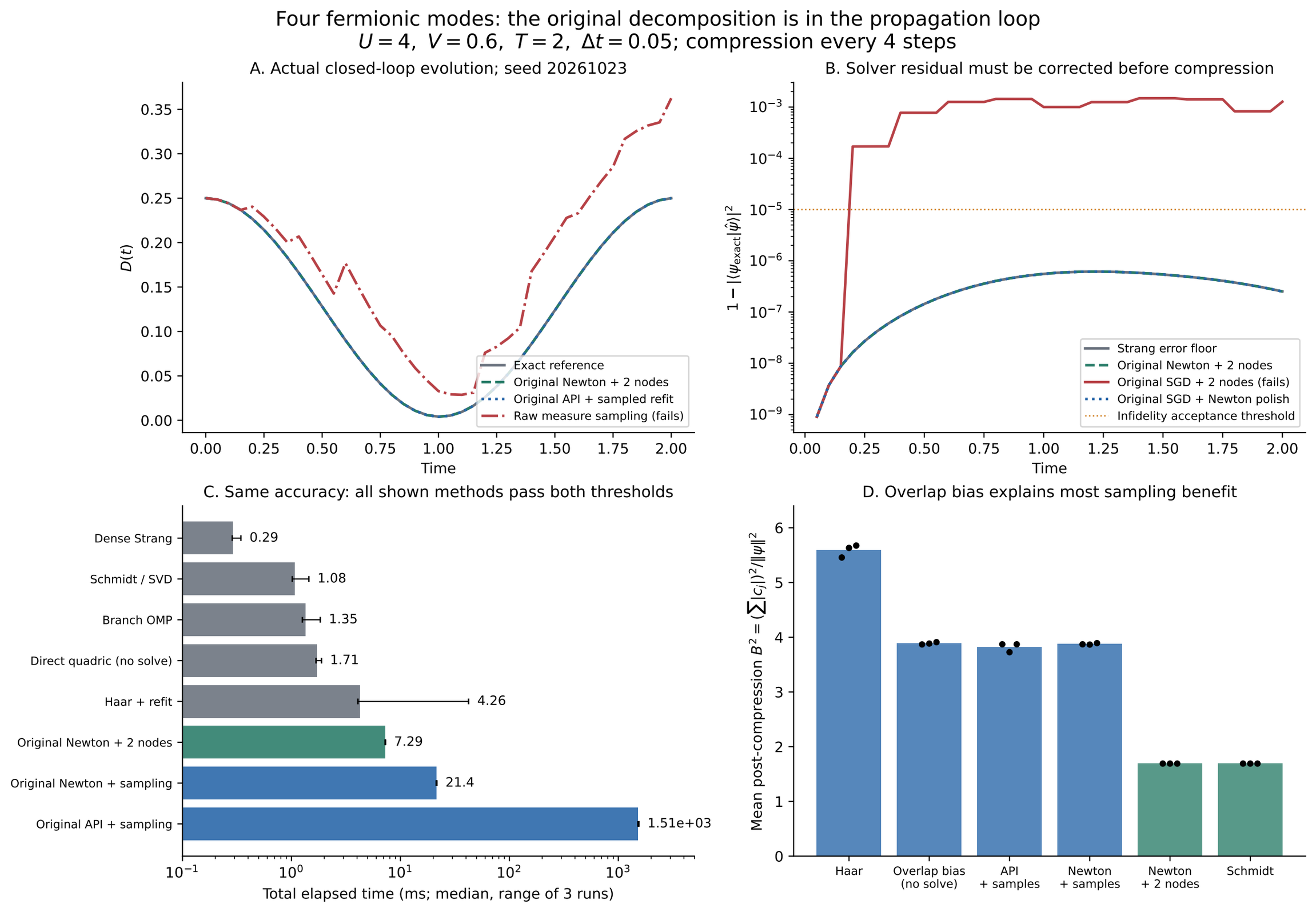
Four fermionic modes: the original decomposition is in the propagation loop
$U = 4,\ V = 0.6,\ T = 2,\ \Delta t = 0.05$; compression every 4 steps
A. Actual closed-loop evolution; seed 20261023
$D(t)$
Time
Exact reference
Original Newton + 2 nodes
Original API + sampled refit
Raw measure sampling (fails)
B. Solver residual must be corrected before compression
$1 - |\langle \psi_{\text{exact}} | \hat{\psi} \rangle|^2$
Time
Strang error floor
Original Newton + 2 nodes
Original SGD + 2 nodes (fails)
Original SGD + Newton polish
Infidelity acceptance threshold
C. Same accuracy: all shown methods pass both thresholds
Dense Strang
Schmidt / SVD
Branch OMP
Direct quadric (no solve)
Haar + refit
Original Newton + 2 nodes
Original Newton + sampling
Original API + sampling
0.29
1.08
1.35
1.71
4.26
7.29
21.4
1.51e+03
Total elapsed time (ms; median, range of 3 runs)
D. Overlap bias explains most sampling benefit
Mean post-compression $B^2 = (\sum |c_j|)^2 / \|\psi\|^2$
Haar
Overlap bias (no solve)
API + samples
Newton + samples
Newton + 2 nodes
Schmidt